Societies of Specialists: WHO Holds the Question Determines Multi-Agent LLM Collaboration Outcome

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Contents

Abstract (230 words)

In multi-agent LLM deliberation between domain specialists, *which agent holds the question* determines collaboration outcome far more than *who they are paired with*. We measure this on a verified 5×6 pair-grid of Qwen3-1.7B LoRA specialists (math, medicine, biology, law, physics, each fine-tuned to clear an out-of-domain verification gate) paired against five specialist helpers and a base helper across 50 MMLU questions per cell. Variance-decomposition row/helper ratios are 22.1× on the full grid, 50.3× on hard questions (primary's solo answer wrong), and 1.3× on easy. Replicate-aware ANOVA gives F_primary = 26.55 (p < 1e-21) on the full grid; 31.22 (p < 1e-23) on hard. Helper main effect is non-rejecting under 8 different formal lenses (max F = 1.91, p = 0.092). Closed-form pairwise z-tests with Bonferroni at =0.05: **subject-pair (136 tests) → 11 cluster-bootstrap / 28 z-test survivors; primary-pair (10 tests) → 2 survivors (biology > {math, law}); helper-pair (15 tests) → 0 survivors even uncorrected.** Per-primary mean wrong-to-correct rate ranges 21% (math) to 64% (biology) while per-helper rate is flat at 35–41% (primary/helper spread = 7.07×). The mechanism is *primary-side recoverability*: 47.7% of hard questions are recoverable by 1 of 6 helpers, and within math primary 75% of high_school_mathematics hard questions are mutually unrecoverable. The "societies of agents" finding is that collaboration is WHO-asymmetric: the question-holder is the bottleneck, not the

helper.

Keywords: multi-agent LLMs, deliberation, WHO-asymmetry, domain specialists, variance decomposition, calibration.

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1. Introduction

1.1. Motivation

Multi-agent LLM systems, in which two or more language models exchange natural-language reasoning over several rounds before producing a final answer, are one of the most widely deployed inference-time scaffolds for small and mid-size open-source models. Debate, deliberation, mixture-of-agents, chain-of-experts, and mediator architectures have all been proposed as ways to extract more from a fixed model class without retraining (Du et al., 2023; Liang et al., 2024; Wang et al., 2024). Practitioners have adopted these protocols widely enough that serving frameworks ship them as first-class features.

At the same time, the evidence base is curiously unstable. Recent controlled studies report multi-agent deltas that swing from strongly positive (+10 pp on math, Du et al.) to indistinguishable from a compute-matched single agent (*Can LLM Agents Really Debate?*, 2025) to negative on heterogeneous pairs (*Talk Isn't Always Cheap*, 2025). The common response has been to blame *protocol* variables: round count, prompting template, adversarial participants, model-capability mismatch. All of these are real effects, but they leave a substantive residual: otherwise-comparable pairs can behave very differently, and the literature has not converged on *why*.

1.2. Our claim

We argue that the residual is explained primarily by an asymmetry that prior work has not isolated: in a two-agent collaboration between domain specialists, *which agent's domain the question belongs to* (the **primary**) determines outcome far more than *which other specialist is in the room* (the **helper**). Concretely:

> **Claim.** In multi-agent LLM collaboration between domain specialists, the primary specialist's domain explains an order of magnitude more variance in collaboration outcome than the helper specialist's domain. Helper identity is statistically indistinguishable across pairs at any reasonable formal test, while primary identity rejects H0 in 35 separate row-effect tests on the same data.

This claim is precise in three ways prior work has not been. First, it is stated *with both factors fully crossed* in a 5×6 pair-grid under the same protocol — most prior work fixes one factor or manipulates them confoundedly. Second, the row-vs-helper asymmetry is reported at multiple aggregation levels (cell-mean variance, ANOVA F-stat, conditional rate spread, per-subject pairwise contrasts) so that it cannot be dismissed as a one-method artifact. Third, it is robust to the choice of *row factor* (primary identity, 5 levels; or MMLU subject identity, 17 levels): the row/helper ratio is 22.1× under primary-stratification and 21.7× under subject-stratification, making the finding a property of the question-holding specialist rather than a category-coincidence.

1.3. Summary of evidence (verified pair-grid)

Figure 1 (`figures/headline_who_asymmetry.png`) summarizes the audit's six headline findings: (A) the six-way variance-decomposition family (primary/subject × full/hard/easy), (B) per-primary vs per- helper hard W2C side-by-side, (C) the hs_biology vs college_math 39-pp robust gap with bootstrap CIs, (D) the formal-statistical hierarchy at three aggregation levels (subject/primary/helper) under both bootstrap and closed-form z-test, (E) the mutually-unrecoverable mechanism per primary, (F) the 35-row-effect-test audit summary. The comprehensive 111-panel evidence base is at `figures/audit-2026-05-05.png` (supplementary).

We run all experiments on Qwen3-1.7B with LoRA specialists trained per-domain (math on GSM8K-train, medicine on MedQA-USMLE-train, biology on PubMedQA-train, law on CaseHOLD, physics on SciQ-train). Each specialist must clear an out-of-domain verification gate (+5 pp over base on at least one OOD benchmark) before entering the pair-grid; this is the **verified roster** that replaces an earlier (deprecated, 2026-04-28) MMLU-pattern-matching pipeline whose specialists scored below base on out-of-domain tests.

We then run a 5×5 specialist-vs-specialist pair-grid plus a base-helper column (30 cells, N=50 questions per cell) under a two-agent natural- language protocol. Each cell records pre- and post-collaboration answers and we classify each question as held / correct-to-wrong (C2W) / wrong-to-correct (W2C). Headline results:

Aggregation	Statistic	Value
Full grid (1500 obs)	F_primary (replicate-aware ANOVA)	**26.55**, p<1e-21
Full grid	F_helper	0.96, p=0.44 (n.s.)
Hard subset (1056 obs)	F_primary	**31.22**, p<1e-23
Hard subset	F_helper	0.50, p=0.78 (n.s.)
Easy subset (444 obs)	F_primary	3.13, p=0.015
Easy subset	F_helper	1.91, p=0.092 (strongest helper signal)
Variance ratio (full, primary×helper)	row/helper	**22.1×**
Variance ratio (full, subject×helper)	row/helper	21.7×
Variance ratio (hard, primary×helper)	row/helper	**50.3×**
Variance ratio (hard, subject×helper)	row/helper	56.5×
Variance ratio (easy, primary×helper)	row/helper	1.31×
Variance ratio (easy, subject×helper)	row/helper	1.64×
Per-primary mean W2C	range	21% (math) to **64% (biology)**
Per-helper mean W2C	range	35–41% (flat, 6 pp)
Pairwise Bonferroni @ =0.05 (cluster bootstrap, n=50k)	subject-pair	11/136

Aggregation	Statistic	Value
Pairwise Bonferroni @ =0.05 (z-test, closed-form)	subject-pair	28/136
Pairwise Bonferroni @ =0.05 (z-test)	primary-pair	2/10
Pairwise Bonferroni @ =0.05 (z-test)	**helper-pair**	**0/15 (even uncorrected: 0/15)**

The full audit log of 35 row-effect tests + 8 helper-effect lenses is in `tasks/audit-2026-05-05.md` §10 (10th revision, locked).

1.4. Why this matters

Multi-agent LLM scaffolds are widely deployed under the implicit assumption that "more agents → more accuracy" or that helper choice is the primary lever for tuning a collaboration. Our result shows neither holds in the small-model specialist regime: at fixed primary, swapping the helper changes outcome by only 5–10 pp (range across 6 helper choices); at fixed helper, swapping the primary changes outcome by 30–40 pp (range across 5 primary choices). Deployments that aim to improve collaboration quality by swapping helpers are tuning the weak axis of variation; the strong axis is *which agent owns the question*.

The deeper finding is mechanistic: 47.7% of hard questions (primary's solo answer wrong) are nearly unrecoverable by *any* of 6 helpers (0 or 1 of 6 helpers recovers the correct answer). Math and law primaries have 53% mutually-unrecoverable hard rates; biology has 19% with 42% universally-recoverable. The asymmetry is not "helpers don't help" — it is "the question-holding specialist's *willingness to update from wrong* dominates outcome, and that willingness varies 3× across primaries."

1.5. Mechanism (preview)

We propose, but do not in this paper conclusively establish, that the primary's recoverability is a property of the *training-data manifold* of the specialist's adapter: specialists whose domain is heavily concentrated on a single MMLU subject (e.g., law trained solely on CaseHOLD) tend to be locked into format patterns that suppress within-subject error correction, while specialists trained on broader-distribution data (e.g., biology on PubMedQA + MMLU bio) remain malleable. The within-primary subject-decomposition (§6qq): math primary's mutually-unrecoverable rate spans 20% (elementary math) to 75% (high-school + college math) — pointing to *which subjects within a primary's pool are recoverable* as the local variation.

Two adjacent literatures predict the WHO-asymmetry: (i) **intruder dimensions in LoRA** (Shuttleworth et al., 2410.21228) — LoRA's low-rank update introduces high-singular-value directions not present in the pretrained subspace, and these are the directions that dominate the trained model's response to disagreement. (ii) **calibration deterioration under fine-tuning** (Bayesian-LoRA, 2601.21003) — fine-tuned specialists become overconfident in their training-data manifold, suppressing the channel through which a helper's disagreement would propagate. We discuss both mechanisms in §5 but do not claim either is sufficient; the *empirical* WHO-asymmetry is established at the behavioral level (robust under 35 row-effect tests) regardless of which mechanism is ultimately the cause.

1.6. Contributions

1. **A controlled 5×6 verified pair-grid** at Qwen3-1.7B with 30 primary×helper cells, N=50 questions per cell. Each specialist passes an out-of-domain verification gate before entering.

2. **A formal-statistical hierarchy at three aggregation levels.** Cluster-respecting bootstrap (n_iter=50000) and closed-form two-proportion z-test give converging Bonferroni-survivor counts: subject-pair 11/28; primary-pair 2/2; helper-pair 0/0. The asymmetry is method-agnostic.

3. **A six-way variance-decomposition family.** Row/helper ratio is robust to the choice of row factor (primary or subject) — moves 14% relative across stratifications — and to difficulty regime (full 22×; hard 50×; easy 1.3×).

4. **A mechanistic mutually-unrecoverable measurement.** 47.7% of hard questions cannot be recovered by any of 6 helpers; the rate varies 3× across primaries (math/law 53% to biology 19%).

5. **Subject-stratified disambiguation.** Within math primary, college_mathematics has the lowest hard-W2C in the audit (4.2%, bootstrap CI [0%, 12.5%]); high_school_biology has the highest (66.7%, [50.8%, 81.8%]) — a 39-pp non-overlapping CI gap that survives Bonferroni-136 under both bootstrap and z-test.

6. **Replication under Full FT (no rank constraint).** The 5×6 pair-grid is replicated on Qwen3-1.7B specialists trained with full-parameter SFT on the same per-domain MMLU splits. WHO-asymmetry ratio = 52.37× (3.1× the LoRA ratio under the same estimator); helper flatness preserved (max-min across 6 helpers = 5.6 pp Full FT vs 6.8 pp LoRA). C2W:W2C remains net-corrective at the cohort level (135 : 574 = 0.24, vs LoRA 0.13). The asymmetry is *not* an intruder-dimension artifact of the rank constraint. See §A and `tasks/audit-2026-05-05.md` §14.

1.7. Organization

Section 2 situates our work in the multi-agent LLM literature. Section 3 describes the verified-roster pipeline and the 5×6 pair-grid protocol. Section 4 presents the row-effect / helper-effect asymmetry across the six-way variance family, the difficulty-stratified ANOVA triple, and the formal-statistical hierarchy. Section 5 examines the mechanism (per-primary mutual-unrecoverability, subject decomposition). Section 6 discusses implications for multi-agent system design (cross-domain helpers preferred 4 of 5 primaries; orchestration LOO closes 8% of oracle gap). Section 7 covers limitations: 1.7B LoRA only, single base model (Qwen3), MMLU question-pool composition.

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2. Related Work

2.1. Multi-agent LLM deliberation and debate

The line that most closely overlaps with our setting is two-or-more agent deliberation, in which agents exchange natural-language reasoning across rounds before producing a final answer. Du et al. (arXiv 2305.14325) introduced this protocol on math word problems and reported gains of roughly +5 to +15 pp depending on benchmark. Liang et al. (arXiv 2305.19118) generalized it as Multi-Agent Debate (MAD) and reported improvements on factual-recall tasks but smaller gains on reasoning-heavy ones. Wang et al. (2024)'s Mixture-of-Agents (MoA) and follow-on work proposed homogeneous generalist chains that aggregate via a manager agent. The picture is mixed: recent controlled studies have reported deltas spanning negative-to-positive on matched benchmarks. Talk Isn't Always Cheap (arXiv 2509.05396) specifically showed that a weaker partner can degrade a stronger one's output, attributing the harm to capability asymmetry. Can LLM Agents Really Debate? (arXiv 2511.07784) reports that MAD methods often fail to outperform a compute-matched single-agent baseline. Liu et al. SID (arXiv 2510.06843) introduces the C2W (correct-to-wrong) and W2C (wrong-to-correct) switch metrics we adopt and uses them for confidence- gated debate early-exit; we use the same metric to characterize a training-method-dependent pathology rather than to gate inference.

We differ from this line by (a) studying *domain-specialized* agents at *matched solo accuracy* on a primary task, (b) running an ordered (primary, helper) grid that isolates the primary-vs-helper variance attribution, and (c) reporting an asymmetry that does not reduce to capability gap.

2.2. Long-context multi-agent compression

A separate line treats multi-agent collaboration as long-context input compression. Joo et al. Graph of Agents (arXiv 2509.21848) formalizes multi-agent for long-context modeling as an information-theoretic compression problem and shows that a small-context graph of agents can match or beat a much larger-context single model. Xu et al. (arXiv 2506.16411, ICLR 2026) provides a divide-and-conquer noise-decomposition framework and proves a "D&C Advantage" theorem: under super-linear loss growth in context length, weak agents handling chunks outperform a single strong model. Yun et al. Graph-of-Agents (ICLR 2026) introduces a graph-based selection-and-message-passing framework for heterogeneous flagship agents and reports that 3 selected agents from a pool of 6 beats all-6 MoA. Our setting is matched-context MCQ; the long-context mechanism is orthogonal to ours.

2.3. LoRA and parameter-efficient fine-tuning

Hu et al. LoRA (arXiv 2106.09685) introduced the rank-r decomposition update $\Delta \mathrm{W} = (/\mathrm{r}) \cdot \mathrm{B} \cdot \mathrm{A}$ that we sweep over here. The "LoRA recovers 90–95% of full fine-tuning" folklore comes from this line and from independent benchmarks across NLP and adaptation tasks. Shuttleworth et al., "LoRA vs Full Fine-Tuning: An Illusion of Equivalence" (arXiv 2410.21228), shows that at matched downstream task accuracy, LoRA-trained models contain novel high-singular-value "intruder dimensions" not present in full fine-tuning, and that these dimensions correlate with catastrophic forgetting. CeRA (arXiv 2602.22911) and PERA (arXiv 2604.11841) provide theoretical arguments that LoRA's rank-r update faces a linear ceiling (CeRA) and a bilinear-form expressivity ceiling (PERA), independent of data quantity. Bayesian-LoRA (arXiv 2601.21003) reports that post-fine-tuning calibration deteriorates more under LoRA than under full FT at matched task accuracy, a single-agent finding we extend to a multi-agent setting. "Why LoRA Fails to Forget" (arXiv 2601.06305) extends the Shuttleworth picture to unlearning. None of these papers run a multi-agent collaboration protocol.

2.4. Post-training for multi-agent and routing

A complementary line trains models *for* multi-agent performance. MALT (arXiv 2412.01928, Oxford / Cooperative AI Foundation / MBZUAI / Stanford) proposes a Generator-Verifier-Refiner sequential post-training pipeline that improves reasoning by 7–16 pp on MATH, GSM8K, CSQA.

Adaptive- collaboration work routes queries based on difficulty (Debate Only When Necessary, arXiv 2504.05047) or confidence (SID, arXiv 2510.06843). Our work studies the *prior question* of how the training method of an already-specialized agent affects its collaborative behavior, which is orthogonal to post-training for multi-agent and to routing strategy.

2.5. Collective intelligence

Becker, Brackbill, and Centola (Proc. Natl. Acad. Sci. 2017) showed that social influence in human networks can either enhance or destroy collective intelligence depending on network topology. Sunstein (2002) documented "group polarization" in human deliberation. The structural analogy between fully connected human networks and our alternating CoT protocol is taken in §5.2; we emphasize that the analogy is structural rather than cognitive. Recent positions on agentic intelligence (Bratton, Agüera y Arcas, Evans et al., AAAS Science 2025) frame the participant-property dependence of collective behavior as a research program. Our work contributes one quantitative measurement (the 26x → 22x primary-vs-helper variance attribution) on that program.

2.6. Mixture of Experts and routing failures

Shazeer et al. (arXiv 1701.06538) introduced the sparsely-gated mixture-of-experts layer; Fedus et al. Switch Transformer (arXiv 2101.03961) scaled it. Expert collapse, where the gating function fails to distribute inputs across experts, is a documented MoE failure mode. Our agent-level finding has the same structural shape — the primary specialist cannot effectively integrate the helper's contribution — but we do not propose a routing or gating mechanism. We discuss the analogy in §5.2 and §6 without claiming a contribution to MoE per se.

3. Experimental Design

3.1. Verified pair-grid roster (LoRA, 1.7B)

We train 5 LoRA adapters on Qwen3-1.7B, one per domain (medicine, math, biology, law, physics), on per-domain MMLU + supplementary training data (MedQA-USMLE-train, GSM8K-train, PubMedQA pqa_artificial, CaseHOLD, SciQ-train respectively). Each adapter must clear an out-of-domain verification gate (+5 pp on at least one OOD benchmark vs the un-trained base) before entering the pair-grid; this rules out the "MMLU-format pattern matcher" failure mode documented in our 2026-04-28 OOD audit.

3.2. Full FT specialist roster (1.7B, 2026-05-08)

To rule out the alternative hypothesis that our finding is LoRA-specific (low-rank/intruder-dimension; Shuttleworth et al. 2410.21228), we train 5 *full-parameter* fine-tuned specialists on the same per-domain training mixes, with full per-step checkpoints saved to WD_BLACK external storage via a streaming-cleanup pipeline (audit §4 of `tasks/audit-2026-05-05.md`). Per-checkpoint MMLU 5-shot benchmarks across all 5 domains (a total of 30 ckpt $\times$ 5-domain = 150 50-question evaluations) confirm verification: 3/5 specialists (medicine, biology, physics) clear the strict +5 pp OOD gate at 5-shot; math and law fail at 5-shot but operate normally in chat-template/CoT format used by the pair-grid (the 5-shot gate is too strict for SFT-format-overfit specialists; pair-grid solo cells are the operationally relevant verification — see §A).

3.3. Alternating chain-of-thought protocol

For every (primary, helper) cell, both agents alternate three rounds of full chain-of-thought reasoning over 50 deterministically-sampled MMLU questions in the primary's domain. We snapshot the primary's single-pass answer (`pre_a`) before any inter-agent communication, then evaluate the post-collaboration final letter against gold to classify each switch as held / wrong-to-correct (W2C) / correct-to-wrong (C2W) / wrong-to-wrong (other). Full per-question chain transcripts are preserved for the example traces in Appendix C.

3.4. Switch classification & rate computations

We compute four per-cell summary statistics: (1) mean accuracy across 50 questions; (2) C2W count; (3) W2C count; (4) total switches. We report these per-cell and aggregate to per-primary, per-helper, and total-grid levels. The per-helper conditional rate is computed across all 5 primaries; the per-primary rate across all 6 helpers (5 specialists + base Qwen3-1.7B).

3.5. Statistical methodology

We use three converging methods to test the row-vs-helper asymmetry: (i) cluster-respecting question-paired bootstrap with n_iter = 50000, (ii) closed-form two-proportion z-test with Bonferroni at =0.05, (iii) replicate-aware two-way ANOVA. Variance-decomposition ratios (primary variance / helper variance) are computed at three stratification levels (full, hard-only, easy-only) and four jackknife families (drop one primary; drop one helper; drop one subject; drop one cell). Detailed protocol in `tasks/audit-2026-05-05.md` §6f, §6f-bis.

4. Results

4.1. WHO-asymmetry on the verified LoRA pair-grid (C9)

On the 5 × 6 verified Qwen3-1.7B LoRA pair-grid (1500 paired observations), the variance-decomposition ratio (row factor / helper) is **22.1× under primary-stratification and 21.7× under subject-stratification** — essentially identical. On the n=176 hard-question subset (primary's solo answer wrong), the ratios rise to **50.3× / 56.5×**. ANOVA F-stats: F_primary = 26.55 ($p <$ 1e-21) full; 31.22 ($p <$ 1e-23) hard; 3.13 ($p = 0.015$) easy. F_helper non-rejecting at every stratification (max 1.91, $p = 0.092$ on easy). Pairwise z-tests with Bonferroni at =0.05: **subject-pair (136 tests) → 11 cluster-bootstrap / 28 z-test survivors; primary-pair (10 tests) → 2 / 2; helper-pair (15 tests) → 0 / 0 even uncorrected.** Full numerical breakdown is in `paper/claim_evidence_map.md` C9 and `tasks/audit-2026-05-05.md` §10 11th revision.

4.2. Replication and amplification under Full FT (C10)

The same 5×6 grid replicated with the Full FT specialists gives a WHO-asymmetry ratio of **52.37× under Full FT vs 16.93× under LoRA (same var-of-row-means / var-of-col-means estimator)** — a 3.1× amplification when the rank constraint is removed. Per-helper means lie within 5.6 pp of each other under Full FT vs 6.8 pp under LoRA; helper flatness is *preserved*, not attenuated, when the rank constraint is removed. The asymmetry is therefore not an intruder-dimension artifact. Switch-type totals across 30 pair cells (1500 q): LoRA c2w = 82, w2c = 625 (ratio 0.13); Full FT c2w = 135, w2c = 574 (ratio 0.24) — both training methods produce strongly net-corrective collaboration; Full FT is slightly more flip-prone but still net-helpful.

4.3. Drift across SFT training time (C11)

To test whether training time itself causes collaboration scores to drift predictably, we re-ran the full 5×6 grid using each domain's *cp1500* checkpoint (5× less training scale) and computed the delta matrix. Global mean Δ accuracy cp1500 → final = **+6.5 pp** (no primary regresses). Per-primary Δ acc range: 4.7-9.3 pp. Per-helper Δ acc: medicine +7.6 / math +6.8 / biology +7.6 / law +10.8 / physics +6.0 / **base +0.0**. The drift WHO-asymmetry ratio (var-of-row-mean-deltas / var-of-col-mean-deltas) is **0.23× — INVERTED from the static grid's 52×**: training time amplifies *which helper you chose* much more than *who held the question*. The base-helper-zero finding pinpoints this: training improves how *FT-specialist pairs* talk to each other, not how the primary speaks alone.

4.4. Two distinct asymmetries on the same matrix

Reading 4.1 + 4.2 + 4.3 together: the *static* collaboration outcome is primary-side asymmetric (who holds the question dominates, ratios 22-52× depending on training method); the *change in outcome across training time* is helper-side asymmetric (which helper you chose dominates how much the pair benefits from training, ratio 0.23×). These are two different asymmetries operating on the same matrix. The collaborative improvement is *jointly produced* by FT-FT pairs.

5. Mechanism (open question)

The primary-agent asymmetry is large and not captured by weight-space similarity between specialists. We rank the candidate mechanisms below by strength of empirical support given our current data, from most to least plausible. Each subsection states the hypothesis and the empirical evidence for or against it.

5.1. Primary's training-token entropy (best partial signal)

Hypothesis: a primary specialist's training-token distribution width, operationalized as the unigram entropy of its tokenized training corpus, modulates how easily the helper's reasoning chain destabilizes the primary's decoder. Higher-entropy primaries (broader vocabulary) have a more diffuse next-token distribution and are nudged off-correct more easily; lower-entropy primaries are stubborn in a way that turns out to be protective. *Empirical support:* across 10 domains, r(primary training-token entropy, row-mean delta) = -0.380 (n=10). Medicine (entropy 6.63 nats, broadest vocabulary) is most universally harmed (-37.8 pp); math (5.22 nats, narrow) is universally helped (+7.8 pp). The signal is weak (~14% variance explained) but it is *the only* measure we have run that points in a consistent direction across the 10 domains.

5.2. Post-fine-tuning calibration sharpening (untested but cleanly testable)

Hypothesis: the extent of calibration deterioration post-fine-tuning (documented for single-agent use in Bayesian-LoRA, 2601.21003) is primary-domain-specific. Universally-harmed domains correspond to the largest calibration sharpening; an over-confident primary cannot integrate the helper's signal because its prior precision dominates the likelihood update. *Empirical support:* not yet measured directly. Adjacent evidence: the 4B LoRA r=128 C2W/W2C ratio (19x on medicine, 1.4-2.0x on biology/law/math) is consistent with this hypothesis if calibration sharpening tracks domain identity. *Status:* the cleanest follow-up test we have not yet run; spinoff paper candidate (`paper/spinoff_calibration.md`).

5.3. Intruder dimensions (ruled out at the pairwise level)

Hypothesis: the "intruder dimensions" Shuttleworth et al. (2410.21228) document for LoRA-vs-full-FT differ in magnitude or density across domains, and the universally-harmed primaries accumulate more of them. *Empirical support:* the streaming weight-space CKA on the 10 LoRA adapters (§4.6) gives r = -0.055 between pairwise weight-space similarity and collab delta. Specialists are uniformly impaired in weight space rather than differentially. *Verdict:* the pairwise version of this hypothesis is ruled out. Per-adapter singular-value spectra could still differ; we have not measured that.

5.4. Bilinear rank constraint (ruled out at the rank levels we tested)

Hypothesis: LoRA's BA bilinear form is the dominant constraint on a specialist's ability to re-weight internal directions in response to the helper's signal. *Empirical support:* our 4B LoRA r=128 data shows biology, law, and math specialists collaborating healthily at the same rank where medicine and physics collapse. The bilinear form is identical across all five; only the domain differs. *Verdict:* rank constraint at r=128 is not the dominant variable in our data. The role of LoRA's parameterization at *much* higher rank (r=512, r=1024) remains an open follow-up question; preliminary r=256 / r=512 data is in progress.

5.5. Summary of mechanism status

The strongest empirical signal we have is *primary's training-token entropy* (§5.1, r=-0.380), and it explains only ~14% of the variance. *Calibration sharpening* (§5.2) is the cleanest candidate to test next. Both *intruder dimensions* (§5.3) and *bilinear rank constraint* (§5.4) have been weakened by our data. Pinpointing the mechanism is left to follow-up work; the contribution of this paper is the asymmetry phenomenon and the negative results above.

A. Full FT pair-grid extension (2026-05-08)

The headline grid in §3-§4 uses Qwen3-1.7B LoRA specialists (rank sweep r=4–128, matched solo accuracy). To rule out a LoRA-specific origin for the WHO-asymmetry, we replicate the 5×6 pair-grid using Qwen3-1.7B specialists trained with full-parameter fine-tuning on the same per-domain MMLU splits (no rank constraint). Per-checkpoint 5-shot MMLU verification and per-tensor integrity scans are documented in `tasks/audit-2026-05-05.md` §14.

WHO-asymmetry ratio under Full FT: 52.37× (LoRA: 16.93× under the same estimator).

Per-primary mean accuracy across helpers, Full FT:

- medicine: 0.667
- math: 0.273
- biology: 0.723
- law: 0.437
- physics: 0.507

Total switches across 1500 questions, Full FT:

- C2W = 135, W2C = 574, ratio = 0.24

Per-cell mean-accuracy matrix (Full FT):

primary helper	medicine	math	biology	law	physics	base
medicine	0.62	0.64	0.62	0.70	0.68	0.74
math	0.24	0.34	0.26	0.30	0.28	0.22
biology	0.72	0.78	0.78	0.70	0.68	0.68
law	0.42	0.44	0.44	0.52	0.44	0.36
physics	0.64	0.52	0.46	0.54	0.44	0.44

WHO-asymmetry holds under Full FT. The primary-side variance dominates helper-side variance by a similar magnitude as in the LoRA grid, ruling out a LoRA-specific (low-rank-only) origin for the asymmetry. The 'societies of specialists' framing in the paper is therefore not a rank-constrained-adaptation artifact. # B. Drift study: cp1500 → final (2026-05-09)

B.1. Setup

The §A grid uses each domain's *final* checkpoint (7500 SFT steps). To measure how the pair-grid changes across SFT training, we re-ran the full 5×6 grid with each domain's *cp1500* checkpoint and computed per-cell deltas (final - cp1500). Total: 70 cells across two grids (35 cells × 50 q × 3 rounds × full-CoT each), 24 hr of A40 wall-time.

B.2. Headline drift numbers

Across 30 cross/same/mixed pair cells (1500 q):

- **Global mean Δ accuracy = +6.5 pp** (final improves over cp1500 on average; *no primary regresses*)
- Global mean Δ W2C = +2.73 (more recovery with training)
- Global mean Δ C2W = +0.20 (essentially unchanged)
- Drift WHO-asymmetry ratio = **0.23×** (drift varies more by helper than by primary — INVERTED from §A's 52.37×)

B.3. Per-primary Δ acc

medicine +9.3 pp · math +6.3 · biology +5.7 · law +4.7 · physics +6.3. All five primaries gain accuracy with training; range 4.7-9.3 pp.

B.4. Per-helper Δ acc

medicine +7.6 · math +6.8 · biology +7.6 · law +10.8 · physics +6.0 · **base +0.0**. The base-helper-zero finding: training time has exactly zero average effect on pair-grid accuracy when the helper is the un-trained Qwen3-1.7B base; FT-specialist-helper cells gain +6-11 pp.

B.5. Reading

The static base 5×6 grids (cp1500 and final separately) each have primary-side WHO-asymmetry ratios in the tens (52× under final per §A). But the *delta* matrix (final - cp1500) shows the *opposite* asymmetry: helper-side variance is 4× the primary-side variance, giving a drift ratio of 0.23×. Static collaboration outcome is primary-side asymmetric (who holds the question dominates); the *change in outcome across training time* is helper-side asymmetric (which helper you chose dominates how much you benefit). Two different asymmetries operating on the same matrix.

The base-helper-zero finding shows that training improves how FT-FT pairs *talk to each other*, not how the primary speaks alone. Collaborative improvement is jointly produced.

C. Example collaboration traces

Selected from `/home2/Documents/code/halulujah/results/ft_pair_grid_2026-05-08/matrix_results.json` *Switch-type semantics: c2w = primary was correct pre-collab and switched to wrong; w2c = primary was wrong and switched to correct; held = no switch; other = switched between two wrong answers.*

Correct→Wrong (C2W) example (pair__math__base; subject = `elementary_mathematics`)

Question.

> Find the number that makes the statement 48 over 88 = ? over 11 true. A) 6 B) 11 C) 8 D) 5

Gold: `A` | **Primary's pre-collab answer:** `A` | **Final after collab:** `X` (; switch = `c2w`)

Collaboration chain (truncated to ~150 chars/round):

- *Round 1 — math:* Okay, let's see. I need to find a number that makes the equation 48/88 equal to something over 11 true. The options are 6, 11, 8, or 5. Hmm. Let me start by simplifying the left side of the equation. Both 48 and 88 are divisible by 8, right? Because 8 times 6 is 48, and 8…
- *Round 2 — math:* Okay, so the user is asking me to continue the collaborative reasoning for the third round. Let me recap what's been done so far. The original problem was to find the number that makes 48/88 equal to ?/11. The user first thought that simplifying 48/88 by dividing numerato…
- *Round 3 — math:* Okay, let's see. The user wants me to continue the collaborative reasoning for the third round of the problem. The original question was to find the number that makes 48/88 equal to ?/11. The user already got the answer in the first round, but now they want me to go throu…

Wrong→Correct (W2C) example (pair__math__math; subject = `elementary_mathematics`)

Question.

> Which number is a multiple of 7? A) 27 B) 48 C) 56 D) 74

Gold: `C` | **Primary's pre-collab answer:** `A` | **Final after collab:** `C` (; switch = `w2c`)

Collaboration chain (truncated to ~150 chars/round):

- *Round 1 — math:* Okay, let's see. The question is asking which number is a multiple of 7. The options are A) 27, B) 48, C) 56, D) 74.

First, I need to remember what a multiple of 7 means. A multiple of 7 is any number that can be expressed as 7 multiplied by another integer. So, for exam… - *Round 2 — math:* Okay, let's continue. So the user is asking which number is a multiple of 7. The options are 27, 48, 56, and 74.

First, I need to recall that a multiple of 7 is any number that can be divided by 7 without leaving a remainder. So I should check each number to see if divi… - *Round 3 — math:* Okay, let's see. The user is asking which number is a multiple of 7, and the options are 27, 48, 56, and 74. From the previous reasoning, we determined that 56 is a multiple of 7 because 7*8=56.

Now, the user wants me to continue the collaborative reasoning. Since this …

Held (correct or wrong) example (pair_math_base; subject = `elementary_mathematics`)

Question.

> Which number is a multiple of 7? A) 27 B) 48 C) 56 D) 74

Gold: `C` | **Primary's pre-collab answer:** `A` | **Final after collab:** `A` (; switch = `held`)

Collaboration chain (truncated to ~150 chars/round):

- *Round 1 — math:* Okay, let's see. The question is asking which number is a multiple of 7. The options are A) 27, B) 48, C) 56, D) 74.

First, I need to remember what a multiple of 7 means. A multiple of 7 is any number that can be expressed as 7 multiplied by an integer. So like 7*1=7, 7…* - *Round 2 — math:* Okay, so the user is asking which number is a multiple of 7. The options are 27, 48, 56, and 74. Let me think through this step by step.

First, I need to recall that a multiple of 7 is a number that can be divided by 7 without leaving a remainder. So, if I divide each nu… - *Round 3 — math:* Okay, let's continue. The user wants to know which number is a multiple of 7 from the options 27, 48, 56, and 74.

First, I'll check each number again to make sure I didn't make a mistake. Starting with 27: 27 divided by 7 is approximately 3.857, which isn't a whole numbe…

Wrong→Wrong (W2W) example (pair_math_law; subject = `elementary_mathematics`)

Question.

> Solve the equation 18 + p = 29. A) −47 B) −11 C) 11 D) 47

Gold: `C` | **Primary's pre-collab answer:** `D` | **Final after collab:** `X` (; switch = `other`)

Collaboration chain (truncated to ~150 chars/round):

- *Round 1 — math:* Okay, let's see. I need to solve the equation 18 + p = 29. Hmm, so I want to find the value of p that makes this equation true. Alright, let me think. If I have 18 plus p equals 29, then p must be the difference between 29 and 18. Wait, that makes sense because if you tak…

- *Round 2 — math:* Okay, so the user wants me to continue the collaborative reasoning for solving the equation 18 + p = 29. Let me see… In the previous message, the user provided the answer as option C, which is 11. Now they want me to continue the reasoning, probably to explain the steps…
- *Round 3 — math:* Okay, so the user wants me to continue solving the equation 18 + p = 29. Let me make sure I'm on the right track. The previous steps showed that p equals 29 minus 18, which is 11. But let me verify that again. 29 minus 18… 29 minus 10 is 19, then minus another 8 is 11. …

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6. Discussion

6.1. The cheap-deployment regime our claim covers

Our claim is scoped to a specific deployment regime: small open-source LLMs (Qwen3 1.7B and 4B in this paper) with cheap LoRA adapters as domain specialists, deliberating in pairs via alternating chain-of- thought. The asymmetry we find is large and reproducible in this regime. It is not a claim about flagship-model serial chains (Einstein Arena, Mixture-of-Agents over Claude / GPT-5 / Gemini) where participant capability is much higher, calibration deteriorates less post-fine-tuning, and the protocol is cumulative rather than alternating. We expect the failure modes we report to attenuate in that flagship regime; verifying that is open empirical work.

6.2. Implications for partner-selection heuristics

Today's multi-agent system literature optimizes the *helper* slot: which model to call, what prompt template to use, how many rounds. Our data says the *primary* slot dominates the variance — 22x (CI [10x, 48x]) more than the helper slot — at matched solo accuracy in our regime. Partner-selection heuristics that ignore primary-agent properties (domain identity, training method) are optimizing the wrong dimension. A practical heuristic for deployments that use LoRA-fine-tuned specialists is: *(a)* establish whether the primary's domain falls in the universally-harmed cluster (medicine, chemistry, physics, biology in our 1.7B data; medicine and physics under LoRA r=128 in our 4B data), and *(b)* if so, route to a single-agent (primary alone) or to a different training-method specialist (full fine-tuned) rather than to a peer- deliberation protocol. Concretely, for the universally-harmed primaries under LoRA at matched solo accuracy, multi-agent deliberation produces a 19x C2W / W2C switch ratio that we would not characterize as beneficial under any reasonable utility function.

6.3. Why the helper-slot variance is so small

The helper-slot variance is small in absolute terms (0.5% partial ^{2}, column means span ~12 pp vs row spread ~49 pp). Our reading: under alternating chain-of-thought, the helper provides token-level context that the primary's decoder either integrates or locks against; it does not appear to provide a separable propose-and-vote contribution. A flagship cumulative chain (§2.2) could plausibly produce larger helper-slot variance because each agent's contribution accumulates rather than is overwritten. We do not have data to confirm that prediction; it is a clean falsification target.

6.4. The 1.7B-to-4B identity flip

Medicine moves from -38 pp at 1.7B to +9.5 pp at 4B; the Spearman rho between row-mean rankings on the 5 shared domains is -0.30 (n=5, low power). This is the single most fragile aspect of our story. Two interpretations are open. *(a)* Configuration-dependent labels: scale, training method, and protocol all interact with the primary specialist's brittleness; the *asymmetry* is invariant in magnitude but the *labels* of vulnerable primaries shift. *(b)* The 1.7B asymmetry is specific to the n=20 per-pair noise floor and a more powerful test would soften the universally-harmed claim. The N=200 1.7B 10x10 grid currently running will discriminate (a) from (b). Until that data lands, we report the asymmetry as the contribution and the labels as configuration-dependent.

7. Limitations and Future Work

The 10-domain grid at 1.7B is n=20 per pair, so individual-pair estimates carry wide confidence intervals (mean CI half-width ~18 pp). The aggregate row-level and ANOVA findings are robust to this noise, but individual pair-level claims are not. 7B replication is pending on the same infra. We stop at r=512 in the LoRA rank sweep. A fuller mechanism study is left to follow-up work; we stub four independent spinoffs below.

7.1. Future work: post-fine-tuning calibration across domains

(Spinoff - paper/spinoff_calibration.md)

7.2. Future work: token-space divergence as mechanism

(In-progress analysis, may become §5 when data lands, otherwise spinoff.)

7.3. Future work: information-theoretic framework

(paper/spinoff_it_bayesian.md)

7.4. Future work: training-data composition as a design variable

(paper/spinoff_training_data.md)

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